HOME COMPUTER
TIM 9919 SOUND GENERATOR KEY FEATURES

3 VOICES WITH 4 OCTAVE MUSICAL RESOLUTION

15 BIT PROGRAMMABLE NOISE SHIFT REGISTER

100 mW AUDIO DRIVE WITH 30DB CONTROL IN 2DB STEPS

8 BIT CPU INTERFACE

5V POWER SUPPLY

I^2L TECHNOLOGY

16 PIN PACKAGE

SOUND GENERATION CONTROLLER

GENERAL DESCRIPTION

The TMS 9919 Sound Generation Controller (SGC) is an I L component designed to provide low cost tone/noise generation capability in microprocessor systems. The SGC is a data bus based I/O peripheral.

KEY FEATURES

* 3 programmable tone generators
* Programmable white noise generator
* Programmable attenuation
* 8 ohm speaker drive capability
* External audio input
* 16 PIN package
* 1 L technology

DEVICE ARCHITECTURE

The device consists of three programmable tone generators, a programmable noise generator, a clock scaler, and an audio summer output buffer. The SGC has a parallel 8 bit interface through which the microprocessor transfers the data which controls the audio output.

TONE GENERATORS

Each tone generator consists of a frequency synthesis section and an attenuation section. The frequency synthesis section requires 10 bits of information (F0-F9) to define half the period of the desired frequency (n). This information is loaded into a 10 stage tone counter, which is decremented at a N/16 rate where N is the input clock frequency. When the tone counter decrements to zero, a borrow signal is produced. This borrow signal toggles the frequency flip-flop and also reloads the tone counter. Thus, the period of the desired frequency is twice the value of the period register. The frequency can be calculated by the following:

$$f = \frac{N}{32n}$$

The output of the frequency flip-flop will feed into a four state attenuator. The attenuator values, along with their bit position in the data word, are shown in table. 14 multiple attenuation control bits may be true simultaneously. Thus, the maximum attenuation is 30 db.

TABLE 14: ATTENUATION CONTROL

BIT POSITION				
A0	A1	A2	A3	WEIGHT
0	0	0	1	2 DB
0	0	1	0	4 DB
0	1	0	0	8 DB
1	0	0	0	16 DB
1	1	1	1	OFF

NOISE GENERATOR

The noise generator consists of a noise source and an attenuator.

The noise source is a 15 stage shift register with an exclusive OR feedback network. The feedback network will have provisions to protect the shift register from getting locked in the zero state.

The feedback network will have two feedback tap configurations as determined by the FB control bit.

TABLE 15: NOISE FEEDBACK CONTROL

FB	CONFIGURATION
0	"Periocic" noise
1	"White" noise

Whenever the FB bit is changed, the shift register is cleared. The shift register will shift at one of four rates as determined by the two NF bits. The fixed shift rates are derived from the input clock.

TABLE 16: NOISE GENERATOR FREQUENCY CONTROL

BITS NF0	NF1	SHIFT RATE
0	0	N/512
0	1	N/1024
1	0	N/2048
1	1	Tone Gen. #3 output

The output of the noise source is fed to an attenuator similar to the tone generator attenuator.

AUDIO SUMMER/OUTPUT BUFFER

The summer is a conventional 1^2L operational amplifier summing circuit. It will sum the three tone generator outputs, noise generator output, and an external audio source. The output buffer will generate up to 100 milliamperes into a 8 ohm load, if all sources are operating at maximum levels. It is assumed that the speaker will be AC coupled to the chip.

CPU/SGC INTERFACE

The microprocessor interfaces with the SGC by means of the 8 data lines and 3 control lines (WE, CE and READY). Each tone generator requires 10 bits of information to select the frequency and 4 bits of information to select the attenuation. A frequency update requires a double byte transfer, while an attenuator update requires a single byte transfer.

CONTROL REGISTERS

The SGC has eight internal registers which are used to control the three tone generators and the noise source. During all data transfers to the SGC, the first byte contains a three bit field which determines the destination control register. The register address codes are shown in Table 17.

TABLE 17: REGISTER ADDRESS FIELD

R0	R1	R2	DESTINATION CONTROL REGISTER
0	0	0	Tone 1 frequency
0	0	1	Tone 1 attenuation
0	1	0	Tone 2 frequency
0	1	1	Tone 2 attenuation
1	0	0	Tone 3 frequency
1	0	1	Tone 3 attenuation
1	1	0	Noise control
1	1	1	Noise attenuation

DATA FORMATS

The formats required to transfer data are shown below.

UPDATE FREQUENCY (2 BYTE TRANSFER)

1	REG ADDR	DATA	0	X	F0	F1	F2	F3	F4	F5
	R0!R1!R2!	F6!F7!F8!F9!								

BIT 0 FIRST BYTE BIT 7 BIT 0 SECOND BITE BIT 7

UPDATED NOISE SOURCE (SINGLE BYTE TRANSFER)

1	REG ADDR			SHIFT
	R0!R1!R2	FB	X	NF0 NF1

BIT 0 BIT 7

UPDATE ATTENUATOR (SINGLE BYTE TRANSFER)

1	REG ADDR	DATA
	R0!R1!R2	A0!A1!A2!A3

BIT 0 BIT 7

HARDWARE DESCRIPTION

POWER CONNECTIONS

The TMS 9919 uses a sigle 5 volt power supply on pin 15, L500 and L501 is used to decouple the power line.

CLOCK

The TMS 9919 uses a 3.58 MHz clock signal on pin 14, derived from the TMS 9918 video display processor. The SN94624 uses the 447.5 KHz GROM clock.

DATA BUS

The TMS 9919 uses an eight bit parallel data bus connected directly to the 99/4 data bus.

SOUND PROCESSOR SELECTION LOGIC

The logic which selects the sound processor is given in figure 30, p.77.

READY

The READY line of the TMS 9919 is directly connected to the system READY line. This wired-OR convention is possible because NAND gate U506 is an open collector.

WE

The WE signal distinguishes between a write to or a read from the TMS 9919.

EXTERNAL SOUND

External sound is fed to pin 9 of the TMS 9919. The 99/4 system uses two sound signals on pin 9.

1. External sound from the cassette interface connector.
2. Sound from the speech unit.

R501 is used to reduce the speech signal to an appropriate level.

SOUND OUTPUT

Sound is available on pin 7. C502 and R511 are used to reduce the bandwidth. C503, C504 and L501 are used for the same purpose.

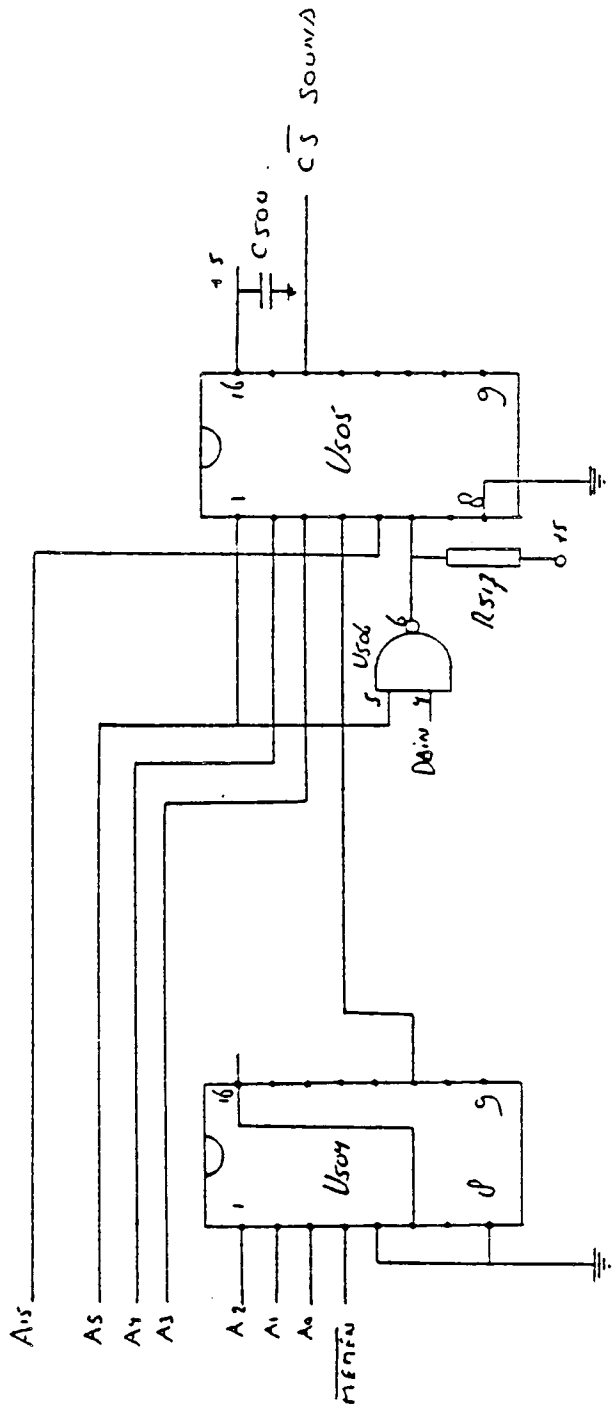


Figure 30. TMS 9919 Sound Processor Selection Logic.

Fig. 30